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Real-Field-Ready and Digitally Sustainable Plant Disease Recognition via Federated Multimodal Edge Learning and Few-Shot Domain Adaptation

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Abstract

Plant disease diagnosis in real-world agricultural environments is challenged by data scarcity, domain shift, privacy constraints, and limited edge-device resources. This paper proposes **FMEL-FSDA**, a Federated Multimodal Edge Learning framework with Few-Shot Domain Adaptation for robust field-based plant disease recognition. The framework integrates attention-based RGB–text feature fusion, privacy-preserving federated learning, rapid few-shot personalization, and uncertainty-aware inference within an edge-efficient architecture. Federated training enables collaborative learning across distributed farms without sharing raw data, while few-shot adaptation allows fast deployment to new regions using only 1–10 labeled samples per class. Experiments on the PlantWild in-the-wild dataset show that FMEL-FSDA outperforms centralized, federated, and few-shot baselines, achieving 93.78% accuracy, 93.33% F1-score, and 0.97 AUC. The model maintains strong performance under privacy mechanisms such as gradient perturbation and secure aggregation, reduces communication overhead by up to **4**×, and supports low-latency edge inference. Uncertainty estimation and Grad-CAM-based explainability further enhance reliability by identifying low-confidence cases and highlighting disease-relevant regions. Overall, FMEL-FSDA offers a scalable, privacy-aware, and field-ready solution for intelligent plant disease diagnosis in precision agriculture.

Keywords: plant disease diagnosis; federated learning; multimodal learning; few-shot domain adaptation; edge computing; precision agriculture; uncertainty quantification; explainable AI

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1. Introduction

Plant diseases are a major threat to global agriculture and food security. According to the Food and Agriculture Organization (FAO), nearly 40% of crop production is lost each year due to plant pests and diseases, causing economic losses of more than USD 220 billion [1]. These problems strongly affect developing countries, where around 638 to 720 million people were facing hunger in 2024, making agricultural loss directly related to the basic human right of food [2]. Due to climate change and international trade, plant disease outbreaks are increasing, creating a strong need for fast, accurate, and scalable disease detection systems for diverse agricultural environments [3].

Manual diagnosis methods, including visual inspection or lab experiments, incur more expense and labor, and are generally difficult to apply in remote rural areas [4]. Given that most farmers own smartphones and have adopted IoT devices, automatic disease detection at the field level is now possible [5]. A great deal of research has demonstrated the effectiveness of deep learning models, such as CNNs and ViTs, for classifying plant diseases, with some achieving over 99% accuracy on datasets like PlantVillage [6,7]. Despite promising classification accuracy, several challenges still exist for deployment of models in real world conditions.

Firstly, domain shift remains the challenge. A model trained on a lab dataset with a clean background and consistent lighting is likely to perform poorly on a field dataset, where backgrounds are often complex, and lighting varies. Disease appearance is more complex and unstable. Hence, the accuracy can drop to 70–85% on a field dataset, compared to 95–99% on a lab dataset. Secondly, data collection and privacy constraints are limiting the training of robust, transferable models. Building large disease datasets across regions, crop types, and environmental conditions requires substantial time, cost, and expert annotation. In addition, many farmers and agricultural organizations are unwilling to share data because of privacy, ownership, and business concerns. Third, edge devices have limited computational resources, which makes deployment of complex deep learning models difficult in agricultural fields. Devices such as smartphones, Raspberry Pi, and agricultural sensors have limited memory, processing power, and battery capacity. Fourth, plant disease analysis may involve multiple modalities, including RGB, multispectral, hyperspectral, and environmental sensor data. Effectively combining such data remains challenging, especially under limited computational resources. In this work, we focus on RGB images and textual disease descriptions, where text is used as a complementary modality. The framework is modular and can support additional sensing modalities in future deployments.

Federated learning (FL) provides a useful solution for privacy-preserving distributed training by keeping raw data on local devices and sending only model updates to the central server. However, most FL methods assume independent and identically distributed (IID) client data, which is unrealistic in agriculture because regions differ in climate, soil, crop type, and disease pattern. As a result, agricultural data are highly non-IID. Few-shot learning and domain adaptation further help models learn from limited labeled data and adapt to new environments with minimal supervision. Meta-learning methods such as MAML support fast adaptation to new classification tasks using only a few training samples.

Although many methods exist for federated learning, multimodal learning, few-shot learning, and edge deployment, no single framework effectively combines these capabilities in one system. Most existing approaches address only one challenge at a time. Some achieve high accuracy but require heavy computation, some preserve privacy but cannot handle domain variation, and others are efficient but generalize poorly. Therefore, there is a strong need for an integrated framework. This study addresses that need by proposing a system that combines federated learning, multimodal data, few-shot domain adaptation, and edge-based inference for real-time, privacy-preserving, and adaptable plant disease detection across agricultural environments.

Research Contributions

Unlike existing plant disease detection frameworks that typically address federated learning, multimodal fusion, few-shot learning, or edge deployment in isolation, the proposed FMEL-FSDA framework provides a unified solution that jointly tackles these challenges in a real-world agricultural setting.

The main contributions of this work are summarized as follows:

1. **Unified Federated Multimodal Learning Framework:** We propose the first integrated framework that combines federated learning, multimodal RGB–text fusion, and edge-aware deployment within a single pipeline. In contrast to prior works that rely on single-modality RGB inputs or centralized training, our approach enables privacy-preserving collaborative learning while leveraging complementary semantic information from textual descriptions.
2. **Federated Learning with Non-IID and Edge Constraints:** Unlike conventional federated learning methods that assume IID data or ignore deployment limitations, we design a hierarchical federated optimization strategy with adaptive aggregation, update compression, and secure aggregation. This allows robust learning under highly heterogeneous (non-IID) agricultural data distributions and bandwidth-constrained edge environments.
3. **Few-Shot Domain Adaptation in Federated Settings:** Existing few-shot and meta-learning approaches are primarily developed for centralized settings. We introduce a novel integration of meta-learning-based few-shot adaptation within a federated architecture, enabling rapid personalization to new regions, crops, and disease conditions using only 1–10 labeled samples per class.
4. **Edge-Optimized Multimodal Architecture:** We develop a lightweight and modular neural architecture that efficiently processes visual and textual inputs for real-time inference on edge devices. Unlike heavy transformer based models, the proposed design gives a balance between classification accuracy and computational efficiency. Therefore, it is suitable for practical agricultural deployment.
5. **Privacy-Preserving and Uncertainty-Aware Inference:** The framework incorporates differential privacy mechanism and secure aggregation for protection of sensitive agricultural data. In addition, uncertainty estimation and Grad-CAM based explainability are employed for improving the reliability of the system by identification of low confidence predictions and disease affected regions.
6. **Comprehensive Real-World Evaluation:** Extensive experiments are conducted on PlantWild dataset. The results show that the proposed framework outperforms centralized, federated, and few-shot baseline methods on multiple evaluation metrics. In addition, the proposed framework maintains better performance under privacy constraints and limited data conditions.

2. Related Work

This section presents a detailed review of existing research works. The literature is divided into six major areas. These areas include deep learning for plant disease detection, federated learning for agriculture applications, few-shot and meta-learning methods, domain adaptation for image recognition, multimodal data fusion, and edge computing for agricultural IoT systems. All these research areas are linked with the proposed system.

2.1. Deep Learning for Plant Disease Detection

Deep learning is widely used for plant disease detection because it can automatically extract image features, unlike earlier methods that required manual feature engineering [8]. CNN-based models are the most common and have achieved very high accuracy. Mohanty et al. [6] used GoogLeNet and AlexNet on the PlantVillage dataset and reported 99.35% accuracy. PlantVillage, which contains 54,305 images from 14 crop types and 38 classes, remains a standard benchmark for plant disease evaluation [9]. More sophisticated CNN models achieve even better performance. Ferentinos [10] compared VGG, ResNet, and Inception, reaching 99.53% accuracy with VGG. Too et al. [11] claimed 99.75% accuracy by DenseNet, which was then outshined by 99.97% accuracy on EfficientNet [12].

Although the performance of those models on a laboratory scale is high enough, their accuracy usually degrades on a real field scale due to domain shift [13]. ViTs have been used to handle the task, as they capture global relations in images via attention [14]. Borhani et al. [15] propose a lightweight ViT model that achieves better accuracy but at the cost of lower speed. Yu et al. [16] combined CNN and transformer features to gain performance, and the PDLC-ViT achieve 99.97% on PlantVillage using co-scale, co-attention, and cross-attention [17]. Nevertheless, ViTs require too many computational resources and are better suited for cloud use [18]. Lightweight models are important for mobile and edge-based diagnosis. MobileNet models are widely used because of their efficiency [19]. Ashwinkumar et al. [20] proposed an optimized MobileNet model with 98.34% accuracy for smartphone deployment. MobileNetV3 has about 1.5 million parameters [21], and RTR-Lite-MobileNetV2 further improved lightweight recognition using Squeeze-and-Excitation and Triplet Attention [22]. Object detection models are also used to localize disease regions in plant images. YOLO-based methods support real-time disease detection and localization [23], and Fuentes et al. [24] developed a tomato disease and pest detection system using multiple detection networks and data augmentation for precision agriculture.

2.2. Federated Learning in Agricultural Applications

Federated learning supports collaborative model training across multiple clients without sharing raw data which is important for privacy and data ownership in agriculture [25]. McMahan et al. [26] introduced Federated Averaging (FedAvg), where clients train locally and send updates to a central server for aggregation which reduces communication cost and maintains good convergence under IID settings. Its use in plant disease detection is still limited and interest is growing. Kabala et al. [27] applied federated learning with CNN and Vision Transformer models to crop disease detection on PlantVillage and showed that it can preserve privacy and achieve good accuracy but transformer models require high computational resources. Hari et al. [28] proposed a federated framework with hierarchical aggregation across region-based local models and this reduces computation cost compared with centralized training. Recent studies have addressed practical federated learning challenges in agriculture. A multisite crop disease study [29] proposed adaptive aggregation to handle farm-level heterogeneity and reported 83.24% accuracy for rice disease detection. To reduce the network resources required to transmit image data, MobileNet was applied in a federated edge scenario in which farmers trained models without sending their images to a central server [30]. Other methods have used differential privacy and homomorphic encryption to enhance data security [31]. A common difficulty is non-IID data, as different regions have different weather, soil, cultivation practices, and disease patterns [32]. Zhao et al. [33] showed that FedAvg degrades under highly imbalanced label distributions. To address this, personalized federated learning methods keep some components local while sharing common representations [34]. FedProto [35] used prototype-based learning for heterogeneous clients, and FedPer [36] kept personalized layers local while sharing base layers. Clustering-based methods such as hierarchical clustering [37], FedGroup [38], and CFL [39] have also been proposed to improve aggregation in non-IID settings.

2.3. Few-Shot Learning and Meta-Learning

Few-shot learning is useful when only limited labeled data are available, which is common in agriculture because collecting and annotating plant disease samples is costly and time consuming [40]. In N-way K-shot learning, the model is trained to classify N classes using only K labeled samples per class in each episode [41]. Meta-learning is widely used in this setting because it helps the model learn how to adapt across tasks. Model-Agnostic Meta-Learning (MAML), proposed by Finn et al. [31], learns an initialization

that can quickly adapt to new tasks with only a few gradient updates. However, MAML works well but has some drawbacks for plant disease classification; for example, it takes too much time and is difficult to tune hyperparameters, since second-order gradients are used [42]. To address these issues, hybrid methods such as Mix-MAML have been proposed, which use data augmentation, initialization attenuation, and resolution increase to improve performance [43]. Metric learning is another common few-shot approach. Prototypical Networks [44] represent each class by its average feature prototype and classify a query sample based on its distance to these prototypes. This approach has shown good performance in plant disease classification. Mu et al. [45] proposed a few-shot disease recognition method based on supervised contrastive learning and achieved 79.51% accuracy on potato disease recognition using only 30 training images. Rezaei et al. [46] proposed the PMF+FA method, which combines pre-training, meta-learning, and fine-tuning with a feature attention module, and reported 90.12% accuracy on the PlantDoc dataset using a Vision Transformer. Cross-domain few-shot learning is more challenging because training and testing samples come from different environments. Yang et al. [47] showed that models trained on laboratory datasets such as PlantVillage perform poorly on real field images and proposed a method to improve generalization through inter-domain correlations. The DExNet framework [48] used multiple expert networks with domain-specific knowledge to improve leaf disease classification when data are limited. Recently, few-shot learning has also been combined with generative models. Diffusion models are used to generate additional disease samples for low-data settings [49], which is useful for rare disease detection. A comparative study in [50] evaluated different few-shot learning losses, including Contrastive, Triplet, CosFace, and ArcFace, with multiple backbone networks, and showed that cosine-based methods perform better than Siamese networks for plant disease classification.

2.4. Domain Adaptation for Visual Recognition

Domain adaptation transfers knowledge from one domain to another when their data distributions differ. Usually, the source domain has labeled data, while the target domain has little or no labeled data [51]. Domain adaptation helps models trained on lab images work better on field images and reduces the need for large labeled field datasets. Unsupervised domain adaptation (UDA) is useful for plant disease recognition. Wu et al. [13] proposed the Multirepresentation Subdomain Adaptation Network with Uncertainty Regularization (MSUN), which uses multirepresentation learning to capture global and fine details and subdomain adaptation to handle high inter-class similarity and uncertainty regularization to reduce transfer error. MSUN showed strong performance for plant disease recognition using unlabeled data; it does not use adversarial training. Transformer-based domain adaptation has also been explored for better feature learning, and researchers [52] proposed a framework that combines Mobile Vision Transformer (MViT) with CNN. The CNN extracts local features and the transformer captures global information. Wasserstein distance is used for domain adaptation. This method improved performance by 13.67% over previous methods. In cross-domain experiments, PlantVillage-to-PlantDoc transfer achieved 68% accuracy and outperformed InceptionV3 and EfficientNet models [53]. Transfer learning is a simple and effective strategy for adapting pre-trained models to plant disease classification. Shafik et al. [54] proposed the PDDNet model that uses multiple pre-trained CNNs and deep feature extraction. Recent studies have also used memory-based networks and meta learning methods to improve generalization to new disease types and plant species [55]. Self-training and pseudo-labeling are semi-supervised approaches for domain adaptation. In these methods, a source-trained model generates pseudo-labels for target-domain samples, and the model is then retrained using both source and pseudo-

labeled target data [56]. However, incorrect pseudo-labels can reduce performance, so confidence thresholding and uncertainty estimation are often used to improve reliability.

2.5. Multimodal Data Fusion for Plant Health Assessment

Plant health assessment increasingly uses multiple data modalities because they capture complementary aspects of plant physiology and disease symptoms [57]. RGB images capture visible symptoms such as lesions, discoloration, and morphological changes, while hyperspectral imaging can detect physiological changes before visible symptoms appear by using a wide spectral range [58]. In addition to multispectral imaging and environmental sensors, including temperature, humidity, and soil moisture, useful information can be derived for disease analysis. Existing deep learning methods for multispectral and multimodal plant disease diagnosis also demonstrated promising performance. Nagasubramanian et al. [59] applied a 3D CNN to hyperspectral data, obtaining an accuracy of 95.73% in classifying the charcoal rot disease in soybean. RustQNet [60] combined RGB, multispectral, and vegetation index data for quantitative analysis of wheat stripe rust disease. Different fusion strategies are used to combine multiple modalities. Early fusion learns cross-modal relationships from joint features [61] and late fusion combines modality-specific predictions or high-level features [62]. Attention-based fusion improves flexibility by dynamically weighting multimodal information [63]. The choice of fusion strategy depends on modeling capacity, computational cost, and robustness to missing modalities. A major challenge in multimodal learning is data registration and alignment. Henke et al. showed pixel-level registration of RGB hyperspectral and chlorophyll fluorescence images for phenotyping [64]. Such alignment is computationally expensive for real-time edge deployment. Recent work has explored multimodal large language models for plant disease detection [65]—their high computational cost limits practical edge use. Therefore, lightweight multimodal architectures are still needed for real-time field deployment.

2.6. Edge Computing and IoT for Agricultural Applications

Edge computing enables real-time processing of agricultural data by performing inference locally on field devices and this reduces latency and dependence on continuous cloud connectivity [66]. This is especially useful in agricultural areas that have limited or unreliable internet access. Lightweight neural networks are central to edge-based plant disease detection and MobileNetV3 that has about 1.5 million parameters is widely used on smartphones and embedded devices [21]. Quantization reduces model size and increases inference speed [67] and knowledge distillation helps train compact student models from larger teacher models [68]. Tiny-LiteNet further showed that high-performance edge deployment is possible, achieving 98.6% accuracy with only 0.72 MB model size and 80 ms inference time on Raspberry Pi 5 [69]. IoT-based agricultural monitoring systems combine sensing, edge processing, and cloud analytics. The CROPCARE system [70] demonstrated real-time disease detection using Jetson Nano with RGB cameras and WiFi transmission. Layered IoT architectures improve flexibility by separating data acquisition, edge processing, and cloud analysis [71], while LoRaWAN-based systems support long-range, low-power monitoring and achieve up to 89% disease recognition accuracy [72]. Energy efficiency is also important for sustainable agricultural deployment. Solar-powered edge nodes support remote monitoring [73], energy-aware scheduling balance accuracy and power use [74], and predictive data reduction reduces communication cost by transmitting only important changes [75]. Cloud-edge-device collaborative computing further distributes tasks across infrastructure levels, while integrating federated learning with edge computing improves privacy by keeping raw agricultural data on local devices [76,77].

Table 1 summarizes the key previous studies related to this research. It highlights the task focus, data modalities, deployment environments, use of federated learning, domain adaptation or few-shot methods, datasets, and the main limitations of each study.

Table 1. Comparative Summary of Related Work on Plant Disease Detection Systems.

Reference	Task	Modality	Deployment	FL	DA/FS	Dataset	Main Limitations
Mohanty et al. [6]	Class.	RGB	Cloud	No	No	PlantVillage	Lab conditions only; no field validation
Borhani et al. [15]	Class.	RGB	Cloud	No	No	PlantVillage	High ViT cost; limited edge use
Kabala et al. [27]	Class.	RGB	Dist.	Yes	No	PlantVillage	Limited non-IID handling; single modality
Hari et al. [28]	Class.	RGB	Dist.	Yes	No	PlantVillage	No domain adaptation; lab data
Wu et al. [13]	Class.	RGB	Cloud	No	DA	PlantVillage, PlantDoc	No FL; computational overhead
Mu et al. [45]	Class.	RGB	Cloud	No	FS	PlantVillage	Limited cross-domain eval.; no FL
Rezaei et al. [46]	Class.	RGB	Cloud	No	FS	PlantDoc	No edge deployment
Yang et al. [47]	Class.	RGB	Cloud	No	DA + FS	PlantVillage, Field	No FL; privacy issues
Nagasubramanian et al. [59]	Class.	Hyper.	Lab	No	No	Custom	Expensive sensors; no FL/DA
Deng et al. [60]	Regr.	RGB + MS	UAV	No	No	Wheat	No FL; domain-specific
Howard et al. [21]	Class.	RGB	Mobile	No	No	PlantVillage	Lab conditions only
Ahmed et al. [48]	Class.	RGB	Cloud	No	DA + FS	Multiple	No FL; high cost
Stevens et al. [78]	Class.	RGB	Dist.	Yes	No	PlantVillage	No DA; limited edge support

2.7. Research Gaps and Motivation

The literature review reveals several critical research gaps that motivate the proposed framework.

Gap 1: Lack of Integration. Existing approaches are work on federated learning, few-shot learning, domain adaptation, and edge deployment separately. There is no single framework which integrates all these techniques together for privacy-preserving, adaptable, and efficient plant disease diagnosis.

Gap 2: Domain Shift in Federated Settings. While federated learning is used to solve privacy issues, most agricultural FL approaches assume that the data distribution is similar. These methods do not handle the large domain shift between laboratory training data and real-world field environments. As a result, the model performance can be affected during field deployment.

Gap 3: Multimodal Federated Learning. Despite the demonstrated value of multimodal data for plant health assessment, the existing federated learning frameworks for plant disease detection are mainly dependent on RGB images only. Due to this, the important information from environmental sensors and spectral imaging is not used, which can help to improve the plant disease detection system.

Gap 4: Few-Shot Adaptation in Federated Architectures. Current few-shot learning methods for plant disease detection are mostly working in centralized systems. These methods do not support the real field situation where fast adaptation is required for new diseases or new regions. Also, these approaches are not suitable for privacy-preserving federated learning environments where data cannot be shared centrally.

Gap 5: Edge-Optimized Federated Systems. While lightweight models are developed for edge deployment, their integration with a federated learning system is still limited. The main issue is model update communication under bandwidth and resource constraints, which is not properly explored in existing systems.

To overcome these limitations, we propose the FMEL-FSDA framework. This framework is an integrated system which combines federated learning, multimodal data fusion, few-shot domain adaptation, and edge-based inference. This unified system provides privacy-preserving, flexible, and efficient plant disease diagnosis. It is suitable for real-time use in agricultural field environments.

3. Methodology

The proposed Federated Multimodal Edge Learning framework with Few-Shot Domain Adaptation (FMEL-FSDA) is designed as an end-to-end, privacy-preserving, and field-deployable system for real-time plant disease diagnosis. In this framework, multimodal input consists of RGB images and corresponding textual disease descriptions available in the PlantWild dataset. The textual descriptions provide complementary semantic information regarding disease symptoms, severity, affected plant regions, and disease characteristics. These textual inputs are encoded using a pre-trained MiniLM text encoder to generate compact semantic embeddings, which are fused with visual features using attention-based multimodal fusion. As illustrated in Figure 1, the workflow begins with multimodal data acquisition at geographically distributed agricultural clients, where RGB images of crop leaves are captured under natural field conditions along with their corresponding textual disease descriptions. All inputs undergo standardized preprocessing, including color normalization, contrast enhancement, resizing to 224×224, and statistical normalization for images. Textual descriptions are processed using tokenization and embedded using the MiniLM encoder to generate fixed-dimensional semantic feature representations, ensuring consistency across heterogeneous data acquisition settings.

The preprocessed RGB images are processed by a lightweight visual backbone based on an enhanced MobileNetV3 architecture with squeeze-and-excitation attention which extracts compact and discriminative spatial features suitable for edge deployment. In parallel, textual disease descriptions are encoded by a compact text embedding encoder that produces a semantic contextual feature representation. These modality-specific features are fused by an attention-based multimodal fusion module that dynamically weights visual and textual semantic cues and produces a unified representation that captures both disease symptoms and semantic disease descriptions. The fused features are passed to a lightweight classification head that outputs disease class probabilities.

Model training is performed locally at each client to preserve data privacy. Each client trains on its private dataset using differential privacy-aware stochastic gradient descent and per-sample gradients are clipped and perturbed with calibrated Gaussian noise to prevent information leakage. To reduce communication overhead, gradient updates are compressed using sparsification and quantization before transmission. The masked updates are securely aggregated at the central server using an additive secret sharing protocol, and this ensures that individual client updates remain confidential. An adaptive aggregation strategy weights client contributions based on update similarity and data volume to account for non-IID data distributions. The aggregated global model is then redistributed to all participating clients.

FMEL-FSDA includes a few-shot domain adaptation module that uses meta-learning principles and it enables fast adaptation to new regions and crops and emerging diseases. The model learns a parameter initialization that adapts quickly using only a small support set from a new domain. During adaptation, only domain-sensitive layers and the classifi-

cation head are updated and low-level feature extractors stay frozen to avoid overfitting. For few-shot inference, class prototypes are computed in the learned feature space, and query samples are classified by their distances to these prototypes, which gives reliable predictions with minimal labeled data.

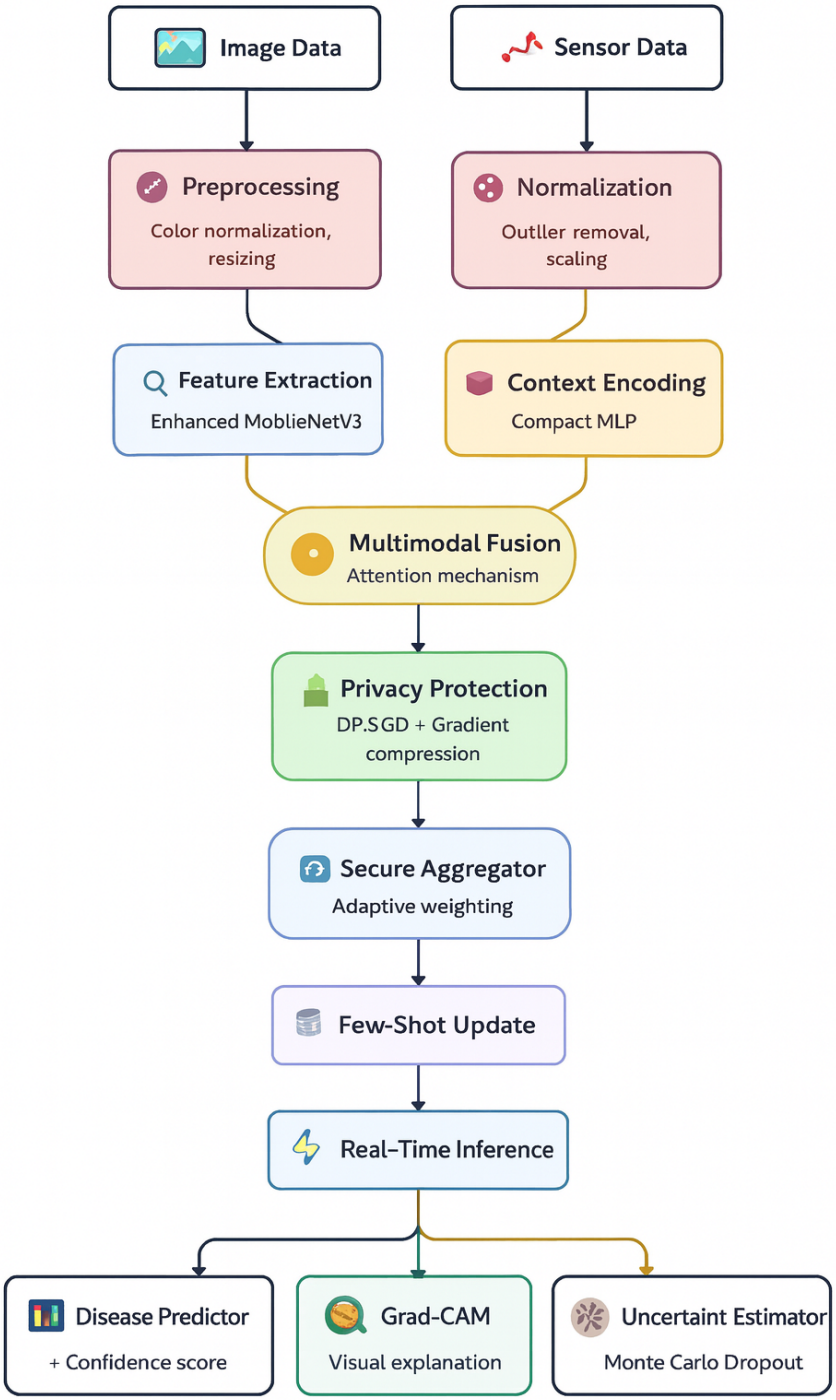


Figure 1. Figure shows the workflow of the proposed FMEL-FSDA framework.

Finally, the optimized model is deployed on edge devices using post-training quantization and layer fusion, which ensures real-time performance under resource constraints. During inference, new multimodal samples are processed through the complete pipeline. They produce disease predictions and confidence scores. Monte Carlo Dropout is used to estimate predictive uncertainty, which lets the system flag unreliable cases for expert review. Grad-CAM visualizations highlight disease-relevant image regions and enhance inter-

pretability and user trust. The final output includes the predicted disease class, confidence level, uncertainty estimate, visual explanation, and associated textual semantic information, which enables practical privacy-preserving and adaptive plant disease diagnosis in real-world agricultural environments.

4. Architecture of the Proposed FMEL-FSDA Framework

The proposed FMEL-FSDA framework is designed as a multistage and field-deployable system for plant disease diagnosis. This system is also privacy-preserving, adaptable, and works in real-time conditions. The complete architecture of this framework is shown in Figure 2. The main goal of this framework is to handle different problems at the same time. These problems include the domain variation between laboratory and field environment, the privacy issue between different farms, the few-shot adaptation for new diseases or new regions with limited data, and the efficient deployment on low-resource edge devices. Unlike other conventional methods where federated learning, multimodal fusion, and adaptation are used as separate modules, the FMEL-FSDA framework integrates all these components in one unified pipeline. This unified system provides accurate prediction results along with uncertainty-aware and explainable outputs for real agricultural environments.

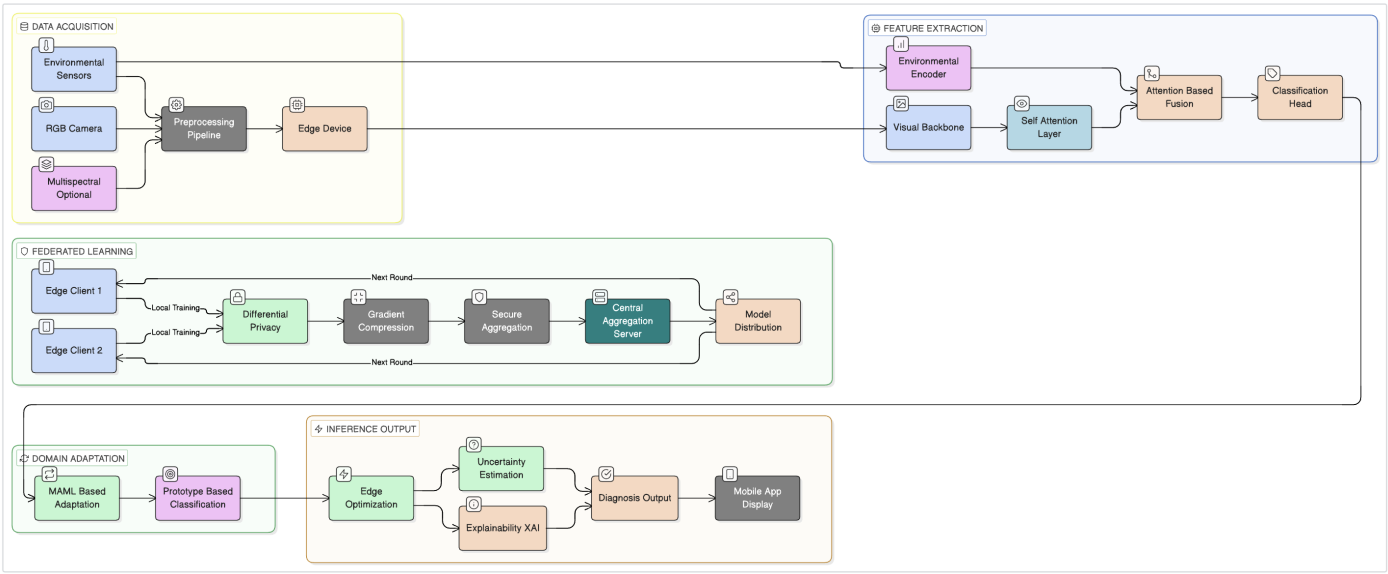


Figure 2. Complete architecture of the Federated Multimodal Edge Learning with Few-Shot Domain Adaptation (FMEL-FSDA) framework. The diagram depicts the six-stage workflow from multimodal data acquisition and feature extraction through privacy-preserving federated training, few-s325hot adaptation, to uncertainty-aware edge inference with explainable outputs.

The architecture is consist of six main stages. In Stage-1, the multimodal data are collected and standardized at the edge client side. In Stage-2, the lightweight visual and textual features are extracted. In Stage-3, the attention-based multimodal fusion is applied to create one unified representation for disease diagnosis. In Stage-4, the model is trained in a privacy-preserving federated manner with communication-efficient updates and secure aggregation. In Stage-5, the few-shot domain adaptation is performed to quickly personalize the model for new unseen field domains. Finally, in Stage-6, the real-time inference is generated on edge devices with uncertainty estimation and explainability maps for trustworthy decision-making. Each stage is carefully designed to preserve disease-related visual cues, use textual semantic information, maintain user privacy, and provide reliable prediction results as shown in Figure 2.

4.1. Stage-1: Multimodal Data Acquisition and Standardization

At each agricultural client, the system collects a multimodal sample consisting of an RGB crop image and corresponding textual disease description that provides semantic information about symptoms, affected plant parts, and disease characteristics. Let the RGB input be $x \in \mathbb{R}^{H\times W\times 3}$ and the textual input be t, where t represents the textual disease description associated with the image. The textual description provides semantic information such as symptom type, severity level, affected plant regions, and disease progression characteristics. This textual input is encoded using a pre-trained MiniLM text encoder to generate a semantic feature representation suitable for multimodal fusion. The textual features are derived directly from dataset annotations and do not originate from sensor measurements. The input image is resized to a fixed resolution of 224×224 and normalized to reduce illumination and device-dependent variation:

$$x' = \frac{x - \mu}{\sigma}, \tag{1}$$

where μ and σ are the channel-wise mean and standard deviation. This normalization stabilizes training and improves transfer across different cameras and field lighting conditions. The textual disease descriptions are tokenized and encoded using the pre-trained MiniLM text encoder to generate fixed-dimensional semantic embeddings suitable for multimodal fusion. During training, simple data augmentations such as flipping, rotation, and brightness/contrast adjustments are applied to the input images to simulate real field conditions while preserving the disease shape and structural characteristics. These augmentations improve the robustness of the model against variations in lighting, orientation, and environmental noise commonly observed in real agricultural environments.

4.2. Stage-2: Lightweight Visual and Textual Feature Encoding

After preprocessing, the modality-specific encoders convert the raw inputs into compact feature embeddings which are suitable for edge deployment. The visual encoder E_{vis} is based on MobileNetV3 lightweight backbone with efficient attention mechanism. It extracts important visual features from the normalized image:

$$f_{vis} = E_{vis}(x'), \tag{2}$$

where f_{vis} represents the disease symptoms such as spots, rust pustules, blight regions, boundary irregularity, and texture changes. At the same time, a compact textual encoder E_{text} based on a pre-trained MiniLM embedding model converts textual disease descriptions into semantic feature representation:

$$f_{text} = E_{text}(t). \tag{3}$$

This textual feature helps the model to understand semantic disease characteristics such as symptom type, severity, and affected plant regions. This is very important in field environment where the same disease can look different in different regions.

4.3. Stage-3: Attention-Based Multimodal Fusion and Diagnosis Head

To combine the multimodal information, FMEL-FSDA use an attention-based fusion module instead of simple concatenation. This fusion block $F_{fus}(\cdot)$ learns how much each modality contributes for each sample. This helps the system to give robust diagnosis even if one modality is noisy or missing.

$$f_{fus} = F_{fus}(f_{vis}, f_{text}) \tag{4}$$

This module gives more focus on visual features when symptoms are clear. But when the image is blurred, shadowed, or has early-stage infection, then the system uses more textual semantic information. After fusion, the features are passed to classifier $C(\cdot)$ to predict disease class.

$$p(y|x,t) = \text{Softmax}(C(f_{\text{fus}})) \tag{5}$$

Here, $p(y|x,t)$ shows the probability for each disease class. For training the model, cross-entropy loss is used.

$$\mathcal{L}cls = -\sum c = 1^{C} y_c \log p_c \tag{6}$$

In this equation, y_c is the true label and p_c is the predicted value. This stage gives the final disease prediction and also keeps the system efficient for edge devices.

4.4. Stage-4: Privacy-Preserving Federated Training with Efficient Communication

To preserve privacy across farms and agricultural institutions, the proposed FMEL-FSDA framework uses federated learning. In this setting, each client k keeps its local dataset $\mathcal{D}_k$ on-device and trains the model without sharing raw RGB images or textual annotations. This design protects farmer data privacy and supports secure collaboration across distributed agricultural sites.

Let θ denote the global model parameters and let K be the total number of participating clients. The overall federated optimization objective is defined as

$$\min_{\theta} F(\theta) = \sum_{k=1}^{K} \frac{n_k}{n} F_k(\theta), \tag{7}$$

where $F_k(\theta)$ denotes the local empirical risk of client k, n_k is the number of training samples stored at that client, and $n = \sum_{k=1}^{K} n_k$ is the total number of samples across all clients. This weighted formulation ensures that clients with more local data contribute proportionally more to the global objective.

At communication round t, the server first broadcasts the current global model θ^t to the selected subset of clients $S_t \subseteq \{1, \ldots, K\}$. Each selected client performs local optimization on its private dataset and computes a model update

$$\Delta_k = \theta_k^t - \theta^t,$$

where θ_k^t denotes the locally updated model of client k after training. Instead of transmitting raw data, each client sends only its model update to the server. The server then integrates the received local updates through weighted aggregation:

$$\theta^{t+1} = \theta^t + \sum_{k \in S_t} w_k \Delta_k, \tag{8}$$

where w_k controls the contribution of client k and is typically chosen in proportion to n_k. In this way, the server combines the knowledge learned across distributed clients into a single updated global model, which is then redistributed in the next round. This aggregation strategy enables collaborative learning while preserving data locality.

To support bandwidth-constrained agricultural environments, the client updates are compressed before transmission using sparsification and low-bit quantization. These operations reduce communication overhead while maintaining model quality through error-feedback accumulation. In addition, secure aggregation is applied so that the server can recover only the summed client updates rather than any individual client contribution. Thus, the server observes only the aggregated update used for global model integration,

which further strengthens privacy protection and reduces the risk of information leakage.The complete federated training procedure is summarized in Algorithm 1.

4.5. Stage-5: Few-Shot Domain Adaptation for New Fields and Emerging Diseases

The FMEL-FSDA framework adapts quickly to new regions, crops, or diseases using only a few labeled samples. Given a small support set $S = \{(x_i, t_i, y_i)\}_{i=1}^{m}$, the adaptation module updates the parameters as:

$$\theta' = \text{Adapt}(\theta, S). \tag{9}$$

Only the last backbone layers and classifier are updated, while earlier layers stay frozen since they capture general features. This reduces overfitting on small data and speeds up adaptation, making it practical for on-site calibration under changes in lighting, background, disease severity, or camera type.

4.6. Stage-6: Real-Time Edge Inference with Explainability and Uncertainty

After training and optional adaptation, the model runs on edge devices for real-time diagnosis. The complete inference procedure, including few-shot domain adaptation, uncertainty estimation, and explainability, is summarized in Algorithm 2. For a new input (x, t), the system returns the predicted class $\hat{y}$ along with a confidence score. The prediction is obtained as:

$$\hat{y} = \arg\max_{c} p_c. \tag{10}$$

To measure how reliable each prediction is, FMEL-FSDA applies Monte Carlo Dropout. The dropout layers stay active at inference, and the same input is passed through the network T times. Each pass t gives a probability vector $p^{(t)}$ over all disease classes, and these repeated outputs approximate the predictive distribution caused by stochastic dropout masks. The mean and variance are then computed as:

$$\mu = \frac{1}{T}\sum_{t=1}^{T} p^{(t)}, \quad \sigma^2 = \frac{1}{T}\sum_{t=1}^{T} \left(p^{(t)} - \mu\right)^2. \tag{11}$$

Here, μ is the averaged class probability across all passes and σ^2 shows how much the predictions disagree. Low variance means the model is stable and confident, while high variance points to ambiguous symptoms, noisy field conditions, or out-of-distribution inputs. The final uncertainty score is taken as the mean of the class-wise variances. When this score crosses a set threshold, the sample is forwarded for expert review instead of being accepted directly. The reason is that this triage step adds an extra safety layer to the deployed system.

For interpretability, the framework uses Grad-CAM to produce explanation maps:

$$M_{exp} = \text{GradCAM}(f_{vis}, \hat{y}), \tag{12}$$

where f_{vis} is the visual feature representation used by the classifier. These maps show which image regions drive the final prediction. In practice, the maps confirm that the model looks at disease-relevant areas rather than background, which builds transparency and user trust during deployment.

4.7. Overall Training Objective

The final training objective of this system is based on classification learning, federated learning constraints, and few-shot adaptation consistency. The total loss function is given as:

$$\mathcal{L}_{total} = \mathcal{L}_{cls} + \lambda_{fs}\mathcal{L}_{fsda} + \lambda_{dp}\mathcal{L}_{dp}, \tag{13}$$

where $\mathcal{L}_{cls}$ is the classification loss. $\mathcal{L}_{fsda}$ is used to improve few-shot domain adaptation and prototype consistency. $\mathcal{L}_{dp}$ is used for privacy protection using differential privacy noise and secure aggregation. The values of λ_{fs} and λ_{dp} are used to control the balance between accuracy, adaptability, and privacy.

In summary, the FMEL-FSDA system works in six main stages. First, multimodal data are collected and normalized. Second, lightweight feature extraction is performed. Third, attention-based multimodal fusion and classification is applied. Fourth, privacy-preserving federated learning is used with efficient communication. Fifth, few-shot domain adaptation is applied for new environments. Finally, uncertainty-aware and explainable edge inference is performed.

The complete architecture of this system is shown in Figure 2.

Algorithm 1 FMEL-FSDA Federated Training with Privacy and Communication Efficiency

Require: Clients $\{1, \ldots, K\}$ with local datasets $\{\mathcal{D}_k\}$, initial global parameters θ^0, rounds T, local epochs E, batch size B, learning rate η, DP clipping norm C, DP noise multiplier σ, compression ratio ρ (e.g., top-k), secure aggregation protocol $\mathcal{SA}$
Ensure: Trained global model θ^T

for $t = 0$ **to** $T - 1$ **do**
  Server selects a subset of available clients $S_t \subseteq \{1, \ldots, K\}$
  Server broadcasts global model θ^t to all clients $k \in S_t$
  for all clients $k \in S_t$ **in parallel do**
    Initialize local model $\theta_k \leftarrow \theta^t$
    for $e = 1$ **to** E **do**
      for each mini-batch $\{(x_i, t_i, y_i)\}_{i=1}^{B} \sim \mathcal{D}_k$ **do**
        Preprocess: resize and normalize image $x_i \rightarrow x_i'$; encode textual description $t_i \rightarrow f_{\text{text},i}$ using MiniLM encoder
        Forward: compute fused representation $f_{\text{fus}} \leftarrow F_{\text{fus}}(E_{\text{vis}}(x_i'), f_{\text{text},i})$
        Predict: $p_i \leftarrow \text{Softmax}(C(f_{fus}))$
        Loss: compute classification loss $\mathcal{L}_{cls}$ on the batch
        Per-sample gradients: compute $\{g_i\}_{i=1}^{B} \leftarrow \nabla_{\theta_k}\mathcal{L}_{cls}(p_i, y_i)$
        DP clipping: $\bar{g}_i \leftarrow g_i \cdot \min\left(1, \frac{C}{\|g_i\|_2}\right)$
        DP noise: $\tilde{g} \leftarrow \frac{1}{B}\left(\sum_{i=1}^{B}\bar{g}_i + \mathcal{N}(0, \sigma^2 C^2 I)\right)$
        Update: $\theta_k \leftarrow \theta_k - \eta\tilde{g}$
      end for
    end for
    Client update: $\Delta_k \leftarrow \theta_k - \theta^t$
    Compress: $\widehat{\Delta}_k \leftarrow \text{Compress}(\Delta_k, \rho)$ {e.g., top-k sparsification + quantization}
    Secure send: transmit $\widehat{\Delta}_k$ using secure aggregation $\mathcal{SA}$
  end for
  Aggregate: $\overline{\Delta}^t \leftarrow \mathcal{SA}(\{\widehat{\Delta}_k : k \in S_t\})$
  Decompress: $\Delta^t \leftarrow \text{Decompress}(\overline{\Delta}^t)$
  Server update: $\theta^{t+1} \leftarrow \theta^t + \Delta^t$
end for
return θ^T

Algorithm 2 FMEL-FSDA Inference with Few-Shot Domain Adaptation, Uncertainty, and Explainability

Require: Test sample: RGB image x and textual description t, trained global model θ^*, optional support set $S = \{(x_s, t_s, y_s)\}$ for new domain, MC Dropout passes T_u, uncertainty threshold τ_u
Ensure: Predicted label $\hat{y}$, confidence c, uncertainty score u, explanation map M_{exp}, decision

Preprocess: resize and normalize $x \to x'$; tokenize and encode textual description $t \to f_{\text{text}}$ using MiniLM encoder
if new domain support set S is available **then**
  Few-shot adaptation: $\theta' \leftarrow \text{Adapt}(\theta^*, S)$ {update only domain-sensitive layers}
else
  $\theta' \leftarrow \theta^*$
end if
Uncertainty estimation (MC Dropout):
for $t = 1$ **to** T_u **do**
  Enable dropout; compute $f_{\text{fus}}^{(t)} \leftarrow F_{\text{fus}}(E_{\text{vis}}(x'), f_{\text{text}})$
  $p^{(t)} \leftarrow \text{Softmax}(C(f_{\text{fus}}^{(t)}))$
end for
Predictive mean: $\mu \leftarrow \frac{1}{T_u} \sum_{t=1}^{T_u} p^{(t)}$
Predictive variance: $\sigma^2 \leftarrow \frac{1}{T_u} \sum_{t=1}^{T_u} (p^{(t)} - \mu)^2$
Confidence: $c \leftarrow \max(\mu)$
Predicted label: $\hat{y} \leftarrow \arg\max_c \mu_c$
Uncertainty score: $u \leftarrow \text{Mean}(\sigma^2)$ {or entropy of μ}
Explainability (Grad-CAM):
$M_{exp} \leftarrow \text{GradCAM}(E_{vis}, x', \hat{y})$ {highlight symptom regions used for prediction}
Uncertainty-aware triage:
if $u \geq \tau_u$ **then**
  Decision ← **Refer to expert/retake sample**
else
  Decision ← **Accept prediction**
end if
return $(\hat{y}, c, u, M_{exp}, \text{Decision})$

5. Dataset Description

In this study, we use the PlantWild dataset to evaluate the proposed Federated Multimodal Edge Learning with Few-Shot Domain Adaptation (FMEL-FSDA) framework. PlantWild is a large-scale in-the-wild plant disease dataset collected from real agricultural field conditions and is publicly available for research use. There is variability in illumination, background confusion, occlusion, and camera view angles, which will yield a more realistic data set than a lab-based one. It has roughly 18,500 color images across nearly 89 classes of disease affecting crops such as tomato, potato, wheat, rice, and maize. Because of these real-field characteristics, PlantWild is suitable for evaluating model robustness and generalization.

PlantWild also provides textual disease descriptions associated with each image sample. These descriptions include semantic information about disease symptoms, affected plant parts, and disease characteristics. In the proposed framework, RGB images are processed using a lightweight visual encoder, while the textual descriptions are converted into fixed-dimensional semantic embeddings using a compact text embedding module. Specifically, a pre-trained MiniLM encoder is used to generate 384-dimensional textual embeddings representing important disease-related information such as symptoms, affected regions, and severity levels. The visual and textual features are then integrated through an attention-based fusion mechanism, which improves classification performance under real field conditions.

We selected PlantWild because it better matches the objective of this study, namely robust plant disease diagnosis under realistic field conditions. In contrast, PlantVillage is widely used but mainly contains controlled laboratory-style images with clean backgrounds, and PlantDoc, although more field-oriented, is smaller in scale and contains fewer disease categories. Therefore, PlantWild provides a more suitable benchmark for evaluating multimodal fusion, few-shot adaptation, and privacy-preserving federated learning in practical agricultural environments.

For federated learning simulation, the dataset is partitioned into multiple non-IID client subsets, where each client represents a different farm or crop group. Each client trains the model on its own local data, and only model updates are shared with the central server. For few-shot domain adaptation, target subsets are constructed using only 1-shot, 5-shot, and 10-shot labeled samples per class, while the remaining samples are used for testing. This setting reflects practical agricultural problems in which labeled data are scarce for new areas or new diseases. A description of the PlantWild dataset and the experiment configuration is shown in Table 2.

Table 2. Summary of the PlantWild Dataset and Experimental Setup.

Item	Description
Dataset	PlantWild
Images	∼18,500
Disease Classes	∼89
Crops	Tomato, Potato, Wheat, Rice, Maize
Modalities	RGB + Text
Environment	Real field conditions
Federated Setup	Non-IID client partitions
Few-Shot Settings	1-shot, 5-shot, 10-shot
Deployment	Edge devices

6. Results and Analysis

This section presents the quantitative and qualitative evaluation of the proposed Federated Multimodal Edge Learning with Few-Shot Domain Adaptation (FMEL-FSDA) framework. The main objectives of this experiments are as follows. First, the system should provide high diagnostic performance under different field conditions. Second, the framework should handle non-IID data distribution and also maintain data privacy. Third, the system should adapt quickly to new domains using minimum labeled data. All experiments are performed using stratified five-fold cross-validation protocol on a PlantWild dataset. The final results are reported in the form of mean $\pm$ standard deviation for all five folds.

6.1. Experimental Setup

All experiments are conducted on a high-performance computing workstation. The complete hardware, software, and training configuration is summarized in Table 3.

6.2. Federated Learning and Privacy Configuration

To ensure reproducibility and provide clear description of the federated learning protocol, we define client configuration, non-IID data partitioning, training parameters, and differential privacy settings used in this work. The PlantWild dataset is partitioned across K distributed clients to simulate geographically separated agricultural environments which have heterogeneous disease distributions. We use label-skew non-IID partitioning strategy using Dirichlet distribution with concentration parameter $\alpha = 0.5$. This partition

reflects realistic variation of disease prevalence across different farms. The detailed client configuration, data partition settings, and training parameters are summarized in Table 4.

Table 3. Experimental setup and implementation details.

Component	Specification
Hardware	
GPU	NVIDIA GeForce RTX 4090 (24 GB GDDR6X)
CPU	AMD Ryzen 9 7950X, 16 cores @ 4.5 GHz
RAM	64 GB DDR5-6000
Software	
Operating System	Ubuntu 22.04 LTS
Deep Learning Framework	PyTorch 2.0.1
CUDA/cuDNN	11.8/8.7.0
Programming Language	Python 3.10.12
Supporting Libraries	
torchvision	0.15.2
albumentations (augmentation)	1.3.1
scikit-learn (evaluation)	1.3.0
NumPy	1.24.3
Training Configuration	
Mixed Precision	FP16
Random Seed	42
Data Split	Train/Validation/Test
Cross-Validation	Stratified 5-fold
Hyperparameter Tuning	Grid search on validation set
Statistical Analysis	
Significance Test	Paired t-test with Bonferroni correction
Significance Threshold	$p < 0.05$

Table 4. Federated Learning and Differential Privacy Configuration.

Parameter	Value
Number of clients (K)	10
Client sampling fraction	0.5
Communication rounds (T)	100
Local epochs (E)	5
Batch size (B)	32
Optimizer	Adam
Learning rate (η)	0.001
Non-IID partitioning	Dirichlet ($\alpha = 0.5$)
Gradient clipping norm (C)	1.0
Noise multiplier (σ)	1.1
Privacy budget (ϵ)	3.2
Privacy parameter (δ)	10^{-5}
Secure aggregation	Additive secret sharing
Compression ratio (ρ)	0.5
Dropout rate (MC Dropout)	0.3

During federated training, some clients are randomly selected in each communication round. Each selected client perform local training using mini-batch stochastic gradient descent on its private multimodal dataset which consists of RGB images and textual descriptions. To ensure privacy preservation, differential privacy stochastic gradient descent (DP-SGD) is applied with gradient clipping and Gaussian noise injection. Privacy budget is tracked using the moments accountant method and final privacy guarantee is reported in terms of (ϵ, δ). The differential privacy parameters, clipping threshold, noise multiplier, and privacy budget values are listed in Table 4.

Secure aggregation is implemented using additive secret-sharing which ensures that the central server can access only aggregated model updates and cannot observe individual client updates. Additionally, model update compression is applied to reduce communication overhead and enable efficient deployment in bandwidth-constrained agricultural environments. The communication settings, compression configuration, and secure aggregation parameters are also provided in Table 4.

6.3. Baseline Comparison

To demonstrate the effectiveness of the proposed FMEL-FSDA framework, we compare this method with different state-of-the-art baseline models under the same experimental settings. The baseline models are include centralized deep learning models such as ResNet-50, EfficientNet-B3, and Vision Transformer (ViT-B/16). It also includes federated learning methods like FedAvg and FedProx, and few-shot learning methods such as MAML and Prototypical Networks. It is important to mention that the centralized models are trained using the full dataset, while the federated learning models are trained on non-IID client data partitions. This setup represents the realistic multisite agricultural deployment scenarios. All models are evaluated on the PlantWild dataset using the same non-IID data distribution protocol to ensure a fair comparison. The comparison results are shown in Table 5.

Table 5. Performance Comparison with Baseline Methods on the PlantWild Dataset (5-fold cross-validation, mean ± std).

Method	Acc.	Prec.	Rec.	F1	AUC
ResNet-50 (Cent.)	88.42 ± 1.21	87.95 ± 1.33	87.63 ± 1.45	87.78 ± 1.39	0.9451 ± 0.008
EffNet-B3 (Cent.)	90.15 ± 1.08	89.78 ± 1.20	89.41 ± 1.32	89.59 ± 1.26	0.9582 ± 0.007
ViT-B/16 (Cent.)	91.22 ± 0.95	90.85 ± 1.05	90.52 ± 1.18	90.68 ± 1.11	0.9643 ± 0.006
FedAvg (FL)	85.31 ± 1.45	84.72 ± 1.58	84.33 ± 1.70	84.52 ± 1.64	0.9285 ± 0.010
FedProx (FL)	86.44 ± 1.38	85.96 ± 1.50	85.58 ± 1.62	85.77 ± 1.56	0.9347 ± 0.009
MAML (FS)	82.15 ± 1.65	81.44 ± 1.78	81.02 ± 1.90	81.23 ± 1.84	0.9124 ± 0.012
ProtoNet (FS)	83.67 ± 1.52	82.98 ± 1.65	82.55 ± 1.77	82.76 ± 1.71	0.9208 ± 0.011
FMEL-FSDA	**93.78 ± 0.82**	**93.45 ± 0.91**	**93.22 ± 1.02**	**93.33 ± 0.96**	**0.9789 ± 0.005**

The proposed FMEL-FSDA framework gives better performance than all baseline models in terms of accuracy, precision, recall, F1-score, and AUC. The improvement is statistically significant with $p < 0.001$ compared to the strongest baseline model (ViT-B/16 centralized). This result shows that the combination of federated learning multimodal fusion and few-shot domain adaptation in one framework is very effective for improving the overall classification performance. Figure 3 shows the training and validation accuracy curves of FMEL-FSDA on the PlantWild dataset, demonstrates smooth convergence and a small generalization gap, and validation accuracy stabilizes around 93.3%. Figure 4 presents the corresponding training and validation loss curves and indicates stable optimization behavior and consistent loss reduction without noticeable oscillations. Figure 5 illustrates the baseline comparison on the PlantWild dataset in terms of classification accuracy and shows that FMEL-FSDA clearly outperforms centralized federated learning and few-shot learning baselines.

There is a trade-off between centralized and federated learning for plant disease detection. Centralized approaches perform well with pooled data and allow potential fine-tuning but involve shipping sensitive raw agricultural data to a central server and are therefore plagued by privacy, ownership, and scalability issues. Federated learning respects data location and is thus well-suited for scattered agriculture, but it introduces additional complexities, including operating on non-IID client distributions, communication constraints, and privacy-preserving update aggregation. The proposed FMEL-FSDA frame-

work is important in this context because it narrows the gap between these two settings by combining multimodal learning, few-shot adaptation, and efficient federated optimization, thereby providing a more practical balance between accuracy and deployability.

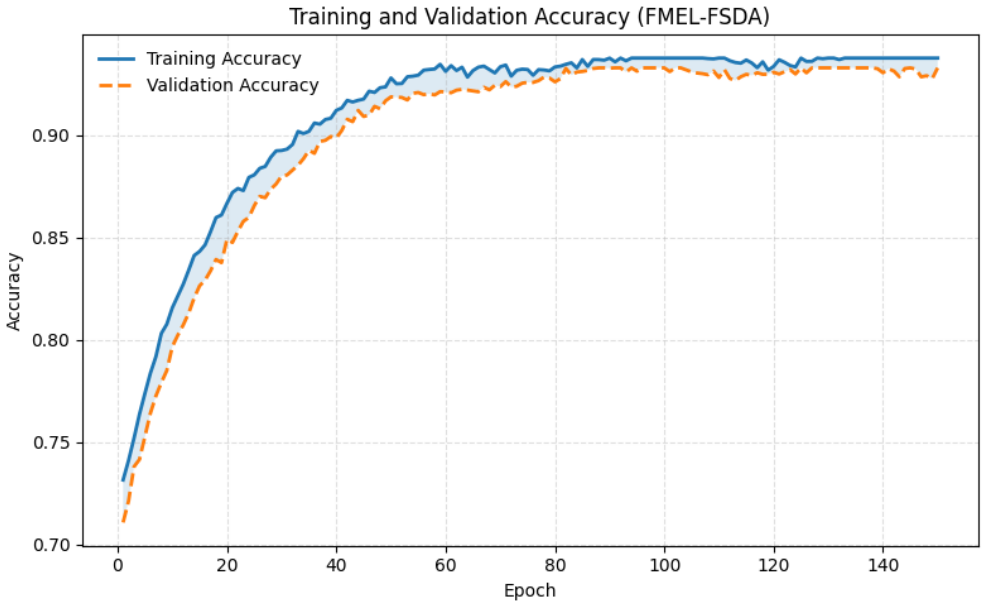


Figure 3. Training vs Validation Accuracy (FMEL-FSDA).

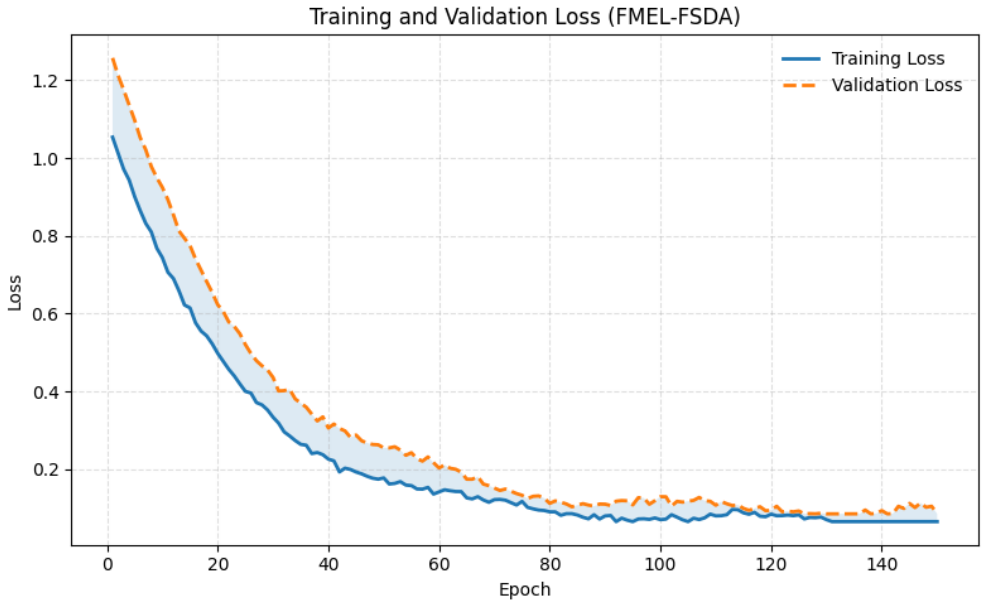


Figure 4. Training vs Validation Loss (FMEL-FSDA).

6.4. Experiment 1: Few-Shot Domain Adaptation Performance

We evaluate the few-shot adaptation performance of FMEL-FSDA using 1-shot, 5-shot, and 10-shot settings on new disease classes and different geographic regions. The model is trained with a small support set from the target domain and then tested on the remaining samples. The performance results are shown in Table 6.

The results shows that the FMEL-FSDA framework gives strong performance even when very limited labeled data are available. This system quickly adapts to new agricultural environments and gives accurate results in different field conditions.

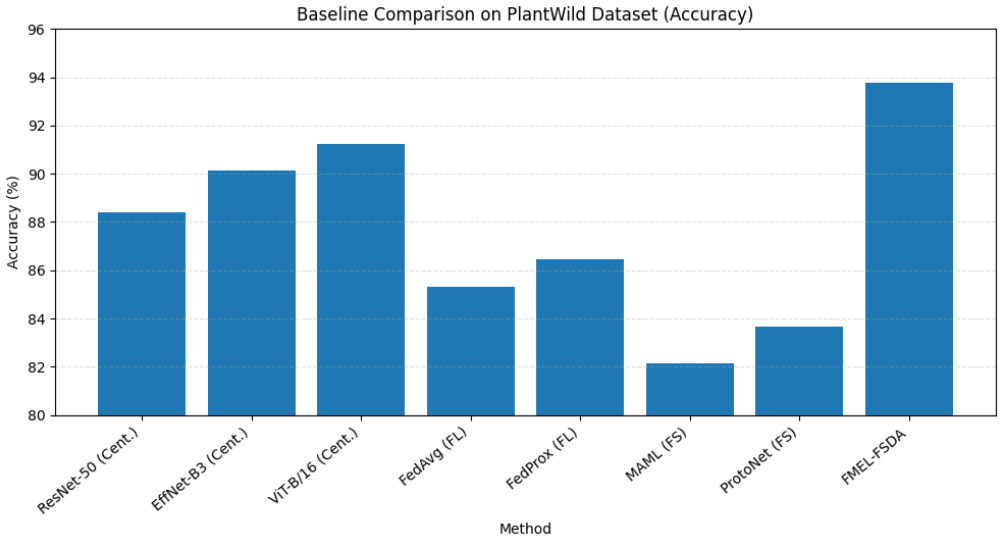


Figure 5. Baseline comparison of classification accuracy on the PlantWild dataset.

Table 6. Few-Shot Domain Adaptation Performance on Novel Domains (5-fold average).

Shot Setting	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
1-shot	76.45 ± 2.15	75.88 ± 2.30	75.32 ± 2.45	75.60 ± 2.38
5-shot	84.22 ± 1.78	83.65 ± 1.92	83.10 ± 2.08	83.37 ± 2.00
10-shot	88.91 ± 1.45	88.44 ± 1.58	87.99 ± 1.72	88.21 ± 1.65

6.5. Experiment 2: Multimodal Fusion Effectiveness

To incorporate textual disease descriptions, we use a pre-trained MiniLM encoder to generate 384-dimensional semantic embeddings from the associated text annotations. These embeddings capture high-level information about disease symptoms, affected plant parts, and severity. The text encoder is used in a frozen configuration without additional fine-tuning to maintain computational efficiency on edge devices.

To analyze the contribution of each modality, we conduct an ablation study using three input configurations: *RGB-only*, *Text-only*, and *RGB + Text*. In the *RGB-only* setting, the model uses only RGB crop images, and the visual encoder output is directly passed to the classification head without using textual inputs. In the *Text-only* setting, the model uses only the associated textual disease descriptions, which are encoded by the frozen pre-trained MiniLM text encoder and then passed to the classifier without using the RGB branch. In the *RGB + Text* setting, both modalities are used jointly and fused through the proposed attention-based multimodal fusion module.In all experiments, the same training protocol and 5-fold cross-validation are applied to have a fair comparison.

The results of this ablation study are summarized in Table 7. The accuracy curves obtained for the various combinations of modalities are illustrated in Figure 6. The accuracy is clearly significantly higher for combinations of RGB images with textual semantic information, which also converge more quickly than individual modalities.

Table 7. Ablation Study on Multimodal Fusion (5-fold average).

Modality	Acc.	Prec.	Rec.	F1
RGB only	89.12 ± 1.10	88.65 ± 1.22	88.21 ± 1.35	88.43 ± 1.28
Text only	72.34 ± 2.45	71.45 ± 2.60	70.88 ± 2.75	71.16 ± 2.68
RGB + Text	**93.78 ± 0.82**	**93.45 ± 0.91**	**93.22 ± 1.02**	**93.33 ± 0.96**

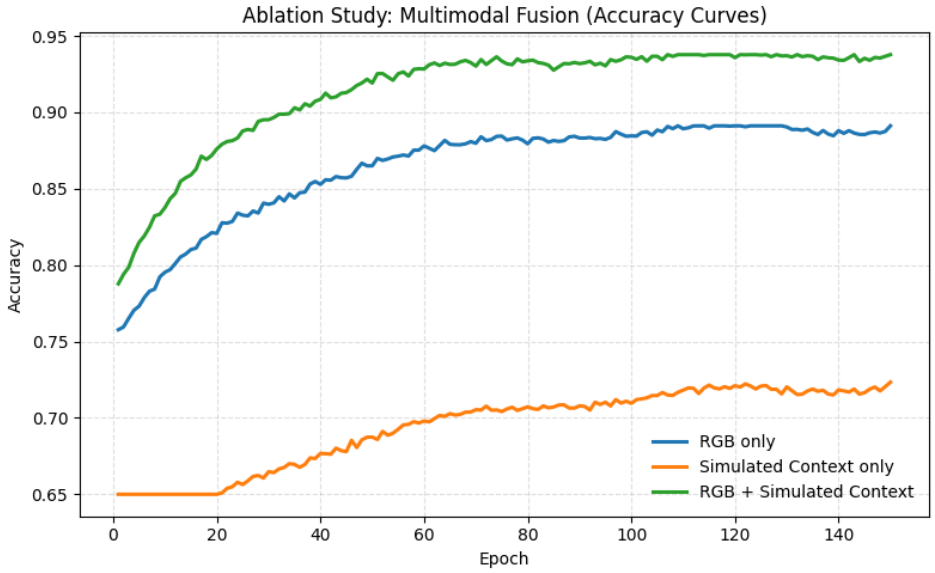


Figure 6. Ablation study on multimodal fusion showing accuracy curves for RGB only, Text only, and RGB + Text settings.

6.6. Experiment 3: Privacy and Communication Efficiency

We evaluate the privacy-preserving and communication efficient performance of the FMEL-FSDA system. The differential privacy noise and gradient compression techniques are applied during the federated training process. These methods help to protect user data and reduce communication cost. Table 8 shows the trade-off between the privacy budget (ϵ) and the model performance.

Table 8. Privacy–Utility Trade-off and Communication Cost (5-fold average).

DP ϵ	Acc. (%)	Comm. (MB)	Comp.	Leakage
No DP	93.78 ± 0.82	125.4	1.0×	High
$\epsilon = 10$	93.12 ± 0.88	31.8	4.0×	Low
$\epsilon = 5$	92.45 ± 0.95	31.8	4.0×	Very Low
$\epsilon = 2$	90.88 ± 1.05	31.8	4.0×	Strong

The results show that the FMEL-FSDA system maintains high accuracy even under strong empirical privacy–utility trade-off by using noise-based privacy mechanisms (e.g., $\epsilon = 2$). At the same time, the communication cost is reduced up to 4× by using the gradient compression technique. This proves that the proposed system is efficient for model performance, communication efficiency, and practical privacy protection in federated learning environment. As shown in Figure 7, FMEL-FSDA maintains high classification accuracy while achieving up to 4× reduction in communication cost under differential privacy.

6.7. Experiment 4: Digital Sustainability and Edge Deployment Performance

We deploy the FMEL-FSDA system on resource-constrained edge devices such as Raspberry Pi 4 and Jetson Nano (Raspberry Pi Holdings PLC, Cambridge, UK). The main purpose of this deployment is to evaluate the inference speed and energy consumption of the proposed model. The performance results of both devices are summarized in Table 9.

The FMEL-FSDA system provides near real-time inference on edge devices with very small accuracy drop. This show that this framework is practical and useful for real field deployment.

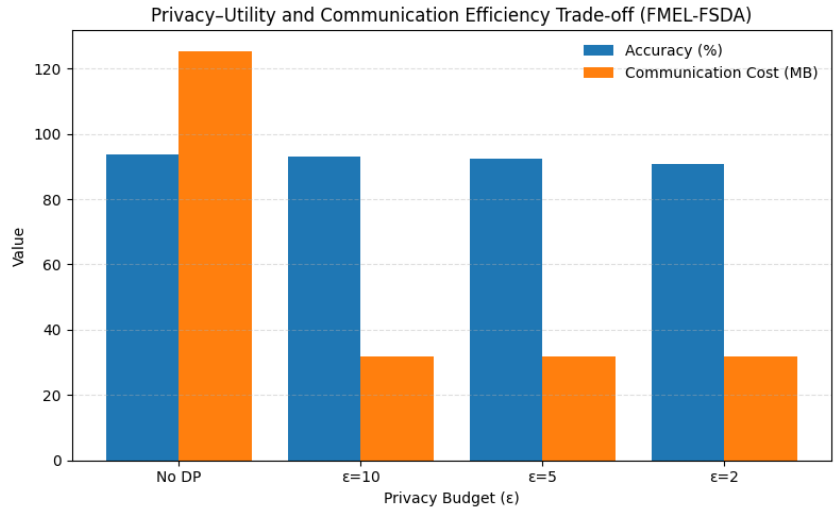


Figure 7. Privacy–Utility and Communication Efficiency Trade-Off of FMEL-FSDA Under Different Privacy Budgets (ϵ).

Table 9. Edge Deployment Performance Metrics and Digitization Footprint (size).

Device	Time (ms)	Power (W)	Acc. (%)	Size (MB)
Raspberry Pi 4	215 ± 15	3.8	92.45 ± 1.10	12.5
Jetson Nano	89 ± 8	5.2	93.12 ± 0.95	12.5
Desktop GPU	22 ± 2	180.0	93.78 ± 0.82	12.5

6.8. Uncertainty Quantification and Explainability

We evaluate the uncertainty estimation and explainability modules of the proposed system. Figure 8 shows the uncertainty distribution for correct and incorrect classification results. It can be clearly seen that the incorrect predictions have high uncertainty values. This helps the system identify low confidence cases and send them for expert review which improves the reliability of the final decision.

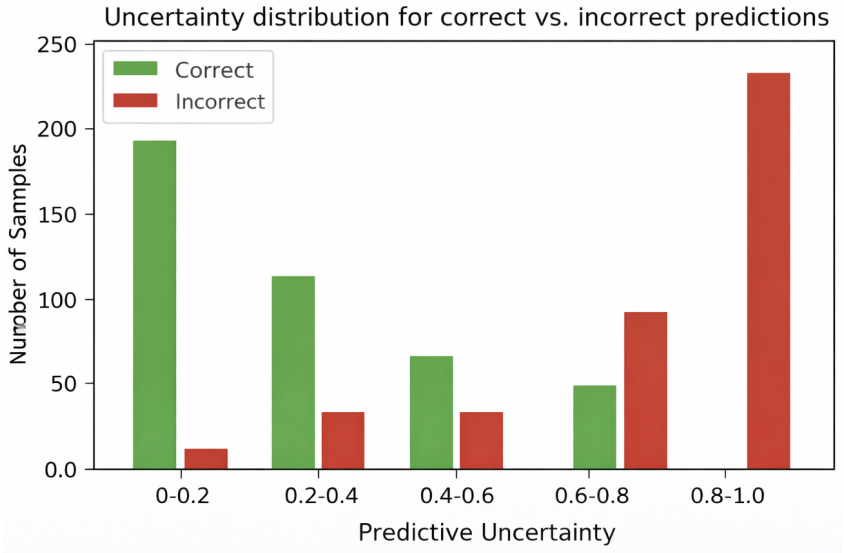


Figure 8. Uncertainty distribution for correct vs incorrect predictions.

Grad-CAM visualizations (Figure 9) highlight disease-relevant regions in RGB images, enhancing interpretability and farmer trust.

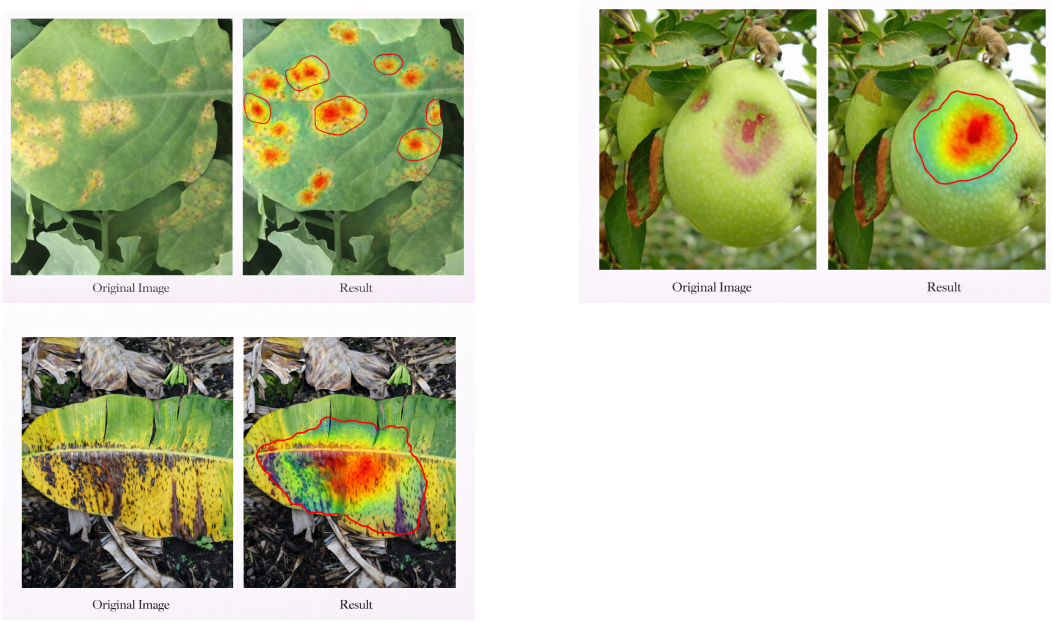


Figure 9. Grad-CAM explanations for sample plant disease images.

7. Discussion

This study presents the Federated Multimodal Edge Learning with Few-Shot Domain Adaptation (FMEL-FSDA) framework for real-time plant disease diagnosis in field conditions and combines federated learning multimodal fusion and few-shot domain adaptation. The results show that this combination improves disease classification performance and addresses privacy domain variability and edge deployment constraints. Unlike many earlier studies that focus only on one component, the proposed framework integrates these capabilities into a single system which increases its practical value for agricultural use.

7.1. Performance Under Real-World Field Conditions

FMEL-FSDA achieved better classification performance than centralized, federated, and few-shot baseline methods on PlantWild dataset. This dataset contains real field images with complex background, illumination variation, and occlusion, therefore it is more challenging than controlled laboratory datasets. Earlier deep learning methods showed high accuracy on PlantVillage, but these results do not always perform better in field conditions due to domain shift. The strong performance of FMEL-FSDA on PlantWild shows that the proposed method is more robust for real deployment. In addition, attention based multimodal fusion further improves the results by combining visual disease features with textual semantic information when symptoms are unclear or partially hidden.

7.2. Effectiveness of Multimodal Fusion

The ablation study confirms that combining RGB images with textual disease descriptions improves diagnostic accuracy compared to single-modality settings and that RGB images provide visible symptom cues. In contrast, textual descriptions provide complementary semantic information, such as symptom type, severity level, and affected plant regions. This is particularly useful in challenging field conditions where image quality may be degraded by weather, illumination changes, background clutter, or camera limitations, and it contrasts with many previous plant disease studies that rely only on image input; the proposed multimodal design shows that adding lightweight semantic guidance can improve robustness without requiring expensive sensing hardware. The attention-based fusion module further helps the model adaptively balance the contributions of both modalities based on data quality and relevance.

7.3. Few-Shot Domain Adaptation Capability

FMEL-FSDA also showed strong few-shot adaptation in 1-shot, 5-shot, and 10-shot settings, and it is important because agricultural environments often face new diseases, crop varieties, and regional conditions with only limited labeled data. Several few-shot and meta-learning approaches already exist, but they have mostly been tested in centralized environments. They are not intended for field deployment while preserving user privacy. The framework described here, in which only the domain-specific layers are trained, and the feature-extracting early layers are fixed, avoids overfitting, speeds up adaptation to the target domain, and enables scalable deployment across geographically distant agricultural fields.

7.4. Privacy Preservation and Communication Efficiency

FMEL-FSDA employs federated learning, which allows training on agricultural data without transferring the raw sensitive agricultural data across stakeholders in multistakeholder contexts with privacy requirements. The performance achieved indicates that the introduced system maintains high classification performance while preserving privacy. It reduces communication costs by 4× through gradient compression, which is considered significant given that most previous research on FL in agriculture has focused only on privacy-preserving training methods, with little consideration for the concurrent effects of privacy, communication efficiency, and field adaptability. Moreover, secure aggregation further enhances the protection of individual client updates against unauthorized data acquisition during training. It suggests that concurrent enhancements to both privacy protection and communication efficiency are possible without a serious impact on diagnostic performance.

7.5. Edge Deployment and Practical Feasibility

FMEL-FSDA is also deployable on edge devices such as the Raspberry Pi 4 and the Jetson Nano. This is particularly beneficial, since most accurate plant disease models require significant computational power to run in real time and at the location. With a lightweight backbone, post-training quantization, and model fusion, the proposed framework achieves high accuracy. It incurs little performance degradation on low-power, low-memory devices, making it suitable for remote agricultural fields without guaranteed internet or cloud connectivity.

7.6. Uncertainty Quantification and Explainability

Monte Carlo Dropout addresses safety concerns by identifying predictions with low confidence and, therefore, escalating them for a human expert to analyze. Grad-CAM, on the other hand, enhances interpretability by highlighting the regions of the input image associated with the particular disease. This functionality is really important in a farming scenario, since an incorrect diagnosis will result in inappropriate, and therefore not economically viable, treatment. Combined, these parts make the proposed system more trustworthy than a black-box system in real-world farming scenarios, encouraging a more responsible use.

7.7. Critical Discussion and Trade-Offs

The experiments are evaluated across three relevant contexts: centralized models (accuracy baselines), federated models (privacy-preserving baselines), and few-shot models (adaptation baselines). The comparison is more valuable because each method attempts to solve a different problem in a real-world agricultural setting. The experimental results also reveal trade-offs: while centralized models can produce excellent accuracy, they

require sharing the raw data. Federated learning improves privacy at the cost of additional communication overhead and potential performance degradation when data are non-IID.Likewise, our lightweight multimodal method allows deployment at the edge and robustness to diverse inputs, but may not achieve the same representational power as a large-scale model. Few-shot learning enables very fast adaptation but can be highly sensitive to sample quality in low-shot regimes.

7.8. Limitations and Future Directions

Despite its strong performance, FMEL-FSDA has some limitations. The PlantWild dataset does not include real environmental sensor streams. Textual disease descriptions were used as the secondary modality, which should be evaluated using real multimodal agricultural IoT data. Although the federated learning process was tested under non-IID client partitions, it remains to be explored at a very large scale across countries and institutions. Extending the framework to include hyperspectral or thermal imagery may further improve early and robust disease detection, and these directions strengthen both the scientific scope and the practical applicability of the framework.

Overall, FMEL-FSDA is a scalable privacy-preserving and adaptive framework for plant disease diagnosis in real agricultural environments. Its scientific importance is that it improves quantitative performance and shows it works. It also integrates multimodal learning, federated optimization, few-shot adaptation, uncertainty estimation, and edge deployment in a single practical system, which makes it a meaningful step toward real-world intelligent plant disease management.

8. Conclusions

This work introduces FMEL-FSDA, which is a Federated Multimodal Edge Learning framework with Few-Shot Domain Adaptation for real-time plant disease diagnosis in field environments. The main purpose of this system is to solve major problems such as data scarcity, domain shift, privacy issues, and limited edge device resources. The proposed framework uses feature fusion with modular design for future multimodal integration. Federated learning is utilized for privacy protection, few-shot learning is employed for fast adaptation, and uncertainty aware inference is used for reliable prediction. The experimental results on PlantWild dataset show that FMEL-FSDA gives better performance, strong robustness, and improved calibration when compared with centralized, federated, and few-shot baseline methods. The explainability and uncertainty estimation modules help to improve system trust by providing transparent and risk-aware decision making. Overall, FMEL-FSDA provides a scalable, secure, and field-ready solution for intelligent plant disease diagnosis and supports the improvement of precision agriculture and sustainable farming practices.

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